

COMP2005 - IIP

Introduction to Image Processing

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COMP2005

- **Lectures**

- Wednesdays 09:0 - 11:00 - LT2

- **Lab**

- Thursdays 09:00 - 11:00 – A32
- **No lab this week**

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Teaching Support Team

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COPM2005 – Learning Outcome

Knowledge and understanding

- Experience implementing programs that manipulate and analyse images – in Python
- Understanding fundamental techniques in image processing and analysis, and their limitations;
- Appreciation of the underlying mathematical principles of the field.

Intellectual Skills

- Apply knowledge of image processing techniques to particular tasks
- Evaluate different techniques in the context of image manipulation and processing.

Professional Skills

- Evaluate the applicability of various algorithms and operators to particular tasks.

Topics Covered

Week No & W/c	Lecture Topic
1 (26/01/26)	Introduction to the module
2 (02/02/26)	Digital Images and Point Processing
3 (09/02/26)	Histogram (Whole Image Understanding)
4 (16/02/26)	Linear and Non-linear Filters
5 (23/02/26)	Binary Images and Thresholding Techniques
6 (02/03/26)	Morphology Image Processing
7 (09/03/26)	Derivatives and Edges + Hough Transform
8 (16/03/26)	Segmentation Techniques
9 (23/03/26)	Metrics for Image Segmentation Evaluation
EASTER HOLIDAYS	
10 (27/04/26)	An Introduction to Image Processing with Deep Learning Techniques ExamSYS Deadline
11 (04/05/26)	Group CW Deadline

COPM2005 - Assessment

- **Group Project (70%)**
 - Python application (group)
 - Explanation of the proposed pipeline & evaluation of results (report)
 - Individual peer assessment form(individual)
- **Due on 06/05/2026 (TBC)**

Coursework

- **In-term ExamSys (30%)**
- **Date of Exam : 30/04/2026 (TBC)**

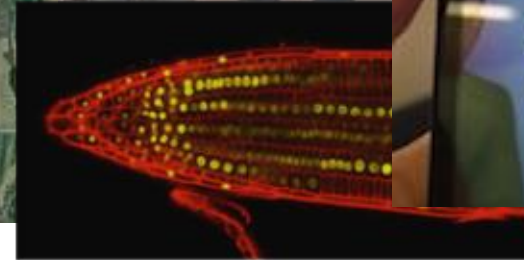
EXAM

Skills needed for Image Processing

- Programming (Python)
- Basic understanding of Maths
- Basic understanding of Algorithmic design
- Physics? Psychology? Physiology?– useful skills but not compulsory
- Ethics
- Critical Analysis and Writing

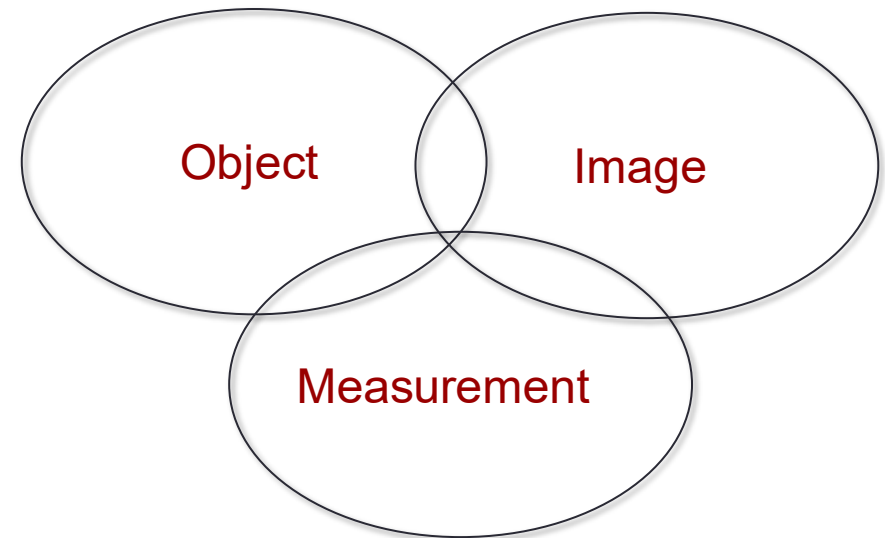
What is Image Processing?

- Collection of topics and techniques related to the use of computers to
 - **Acquire**
 - **Store**
 - **Manipulate**
 - **Model**
 - **Analyse/Interpret**
 - **Display**
 - **Compress**
- Generic techniques that are applied to images from most sources



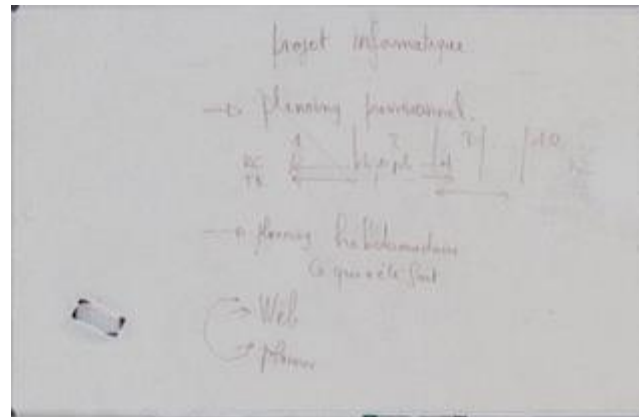
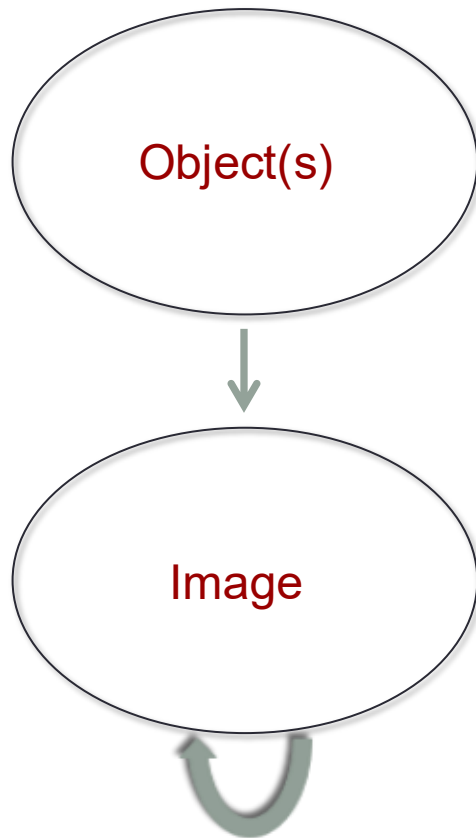
Know Your Limitations

- Four terms are often used together, and some are sometimes confused
 - Image Processing
 - Image Analysis
 - Computer Vision
 - Computer Graphics
- All share representations, underlying mathematics and some algorithms
- Their goals are very different



This is an **Image Processing** module,
with some **Image Analysis**

Image Processing



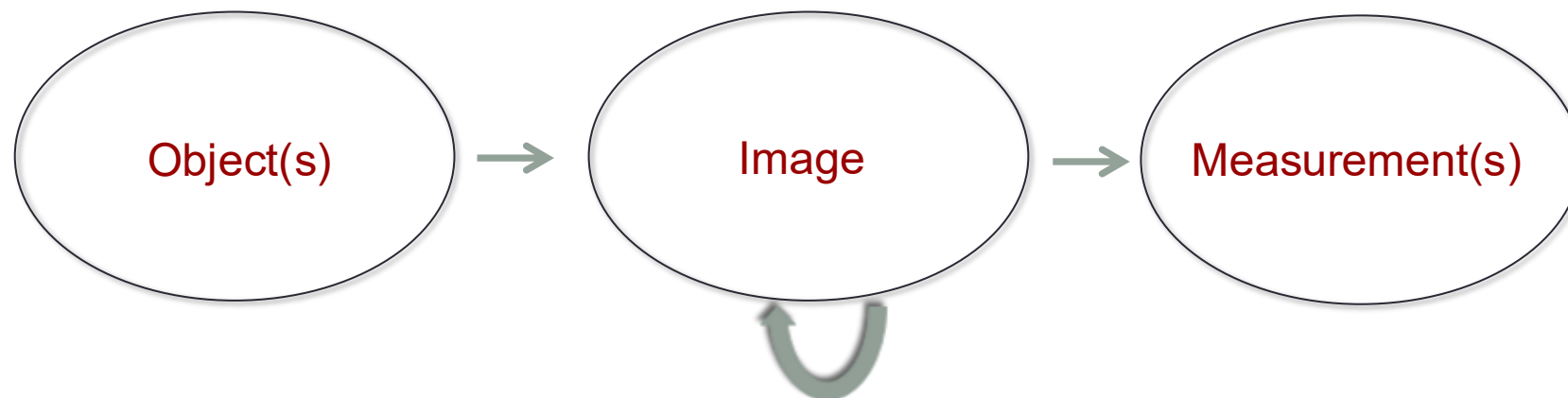
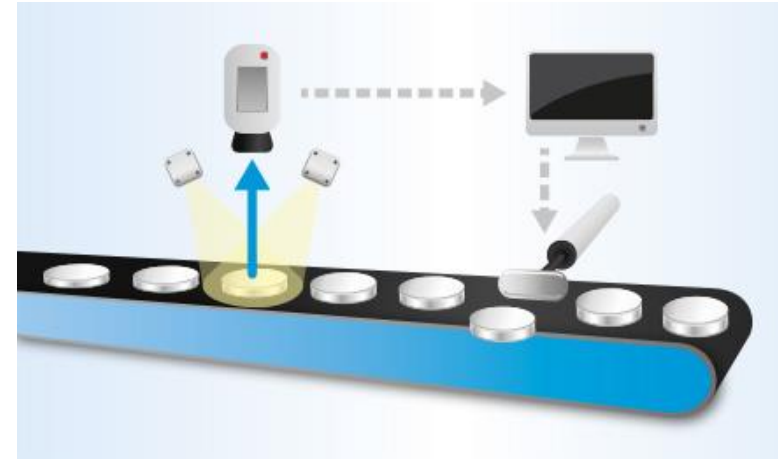
- **Image in, Image out**
- Key information more easily seen/extracted
- More aesthetically pleasing

Image Analysis

- Concerned with making **quantitative** measurements on images:

Pixel in >> Pixels & Features out

- Image acquisition is constrained so that image measurements are a proxy for some real-world value
- Sits between image processing and computer vision



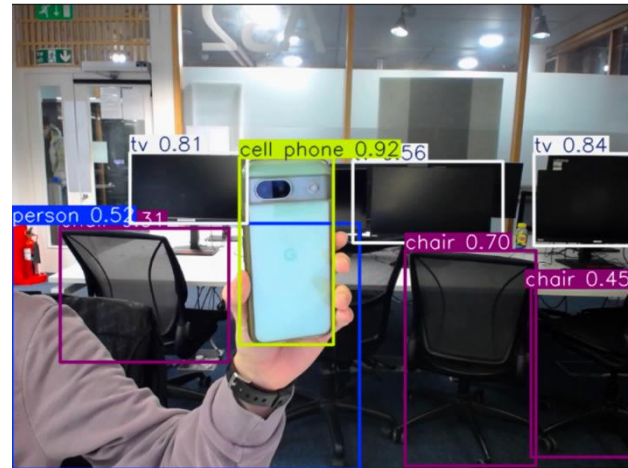
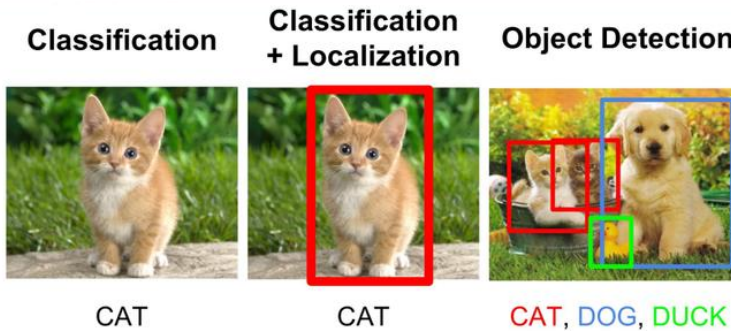
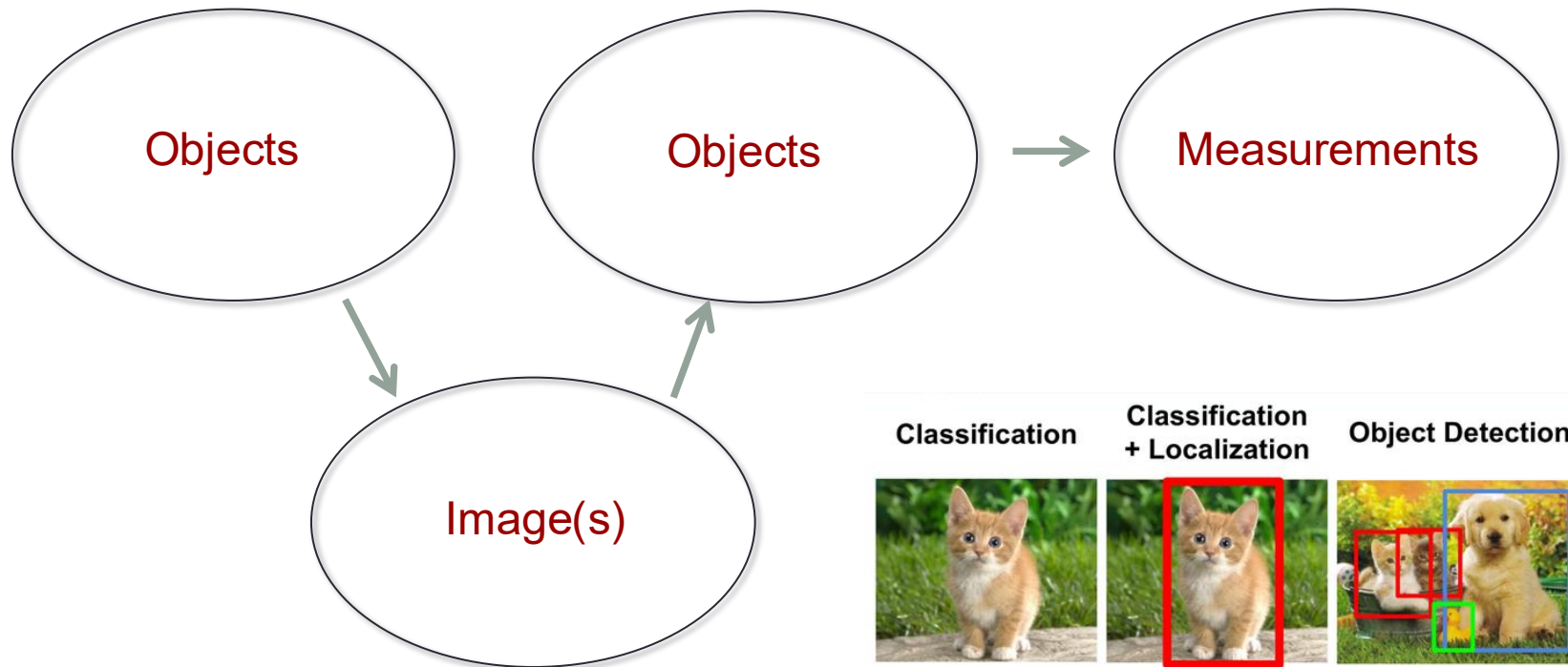
It's not Computer Vision

Computer Vision is concerned with the computational processes that allow representations of the viewed environment to be recovered from images

- **Image in >>> Information out (3D Object, labels, measurements)**



**3D reconstruction
from 2D image**



Object Tracking and Recognition

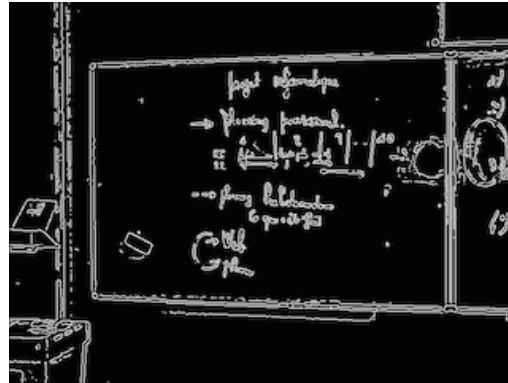
Its not Computer Graphics

3D Objects in >>> 2D Image out

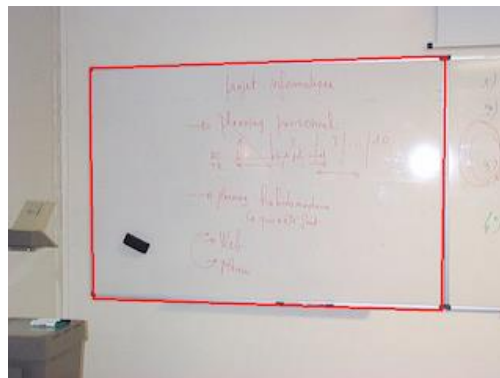
- Focus is on creating images from object models:
 - Lighting and shading modeling
 - Volume modeling
 - Curve and surface modeling
 - Visibility modeling
 - Texture synthesis
 - Character animation
 - Modeling terrain, liquids, fire/smoke, cloth, hair/fur, feathers, skin etc



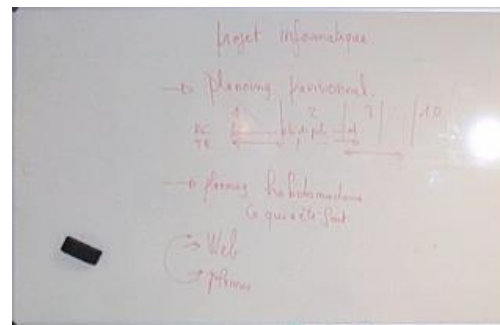
Its a toolkit



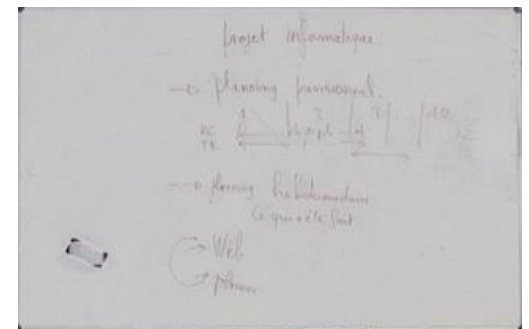
Edge detection



Line finding



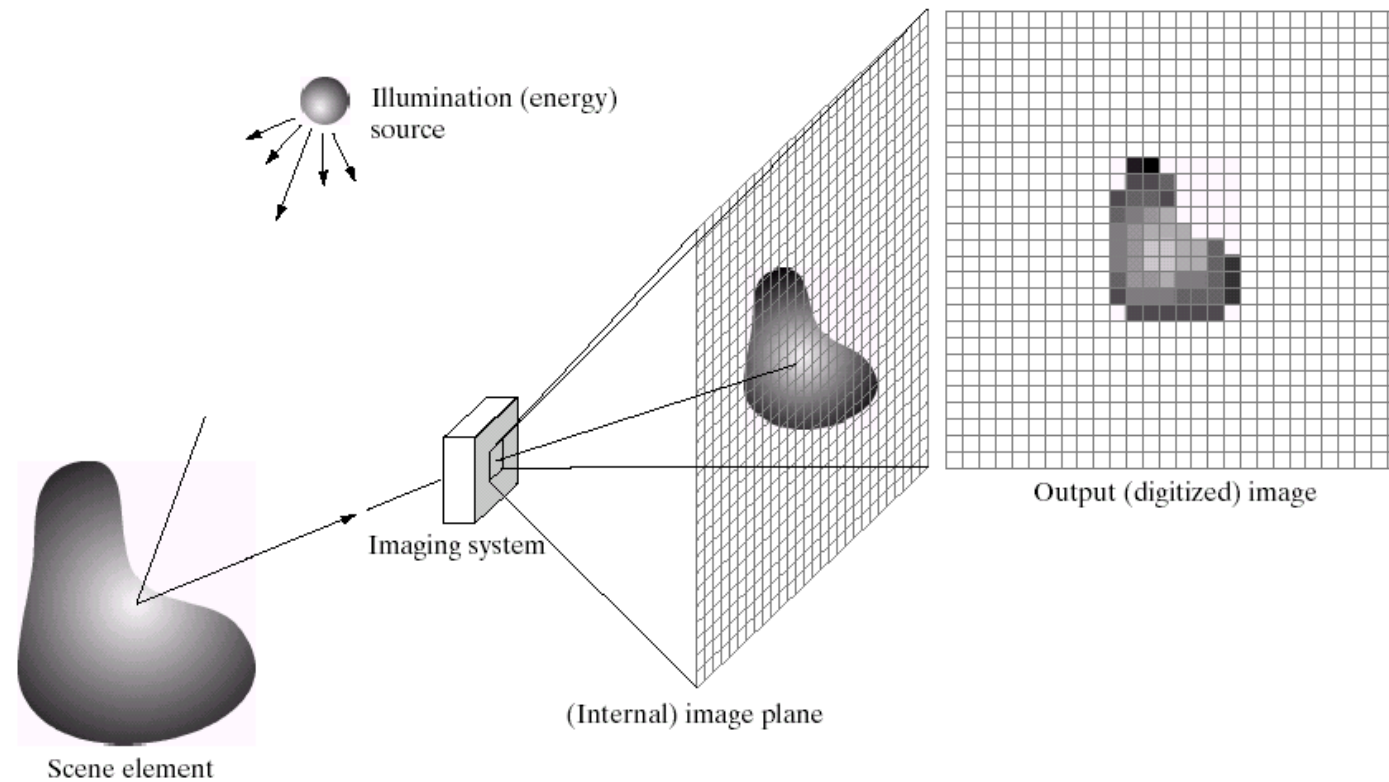
Distortion correction



Illumination correction

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- Image formation, acquisition, representing colour



Digital Images

- A common, **low-level** representation of the viewed world



123	33	234	45
67	90	12	134
56	89	54	67
111	56	67	90
34	...		



Pixel values represent the brightness and colour of the viewed objects, but give no indication of what object, e.g., books, monitors, these numbers refer to – hence low-level

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- Redundancy & image compression – **efficiently** represent image data for storage (minimise disk space) and communication (minimise network bandwidth)



245,760 bytes



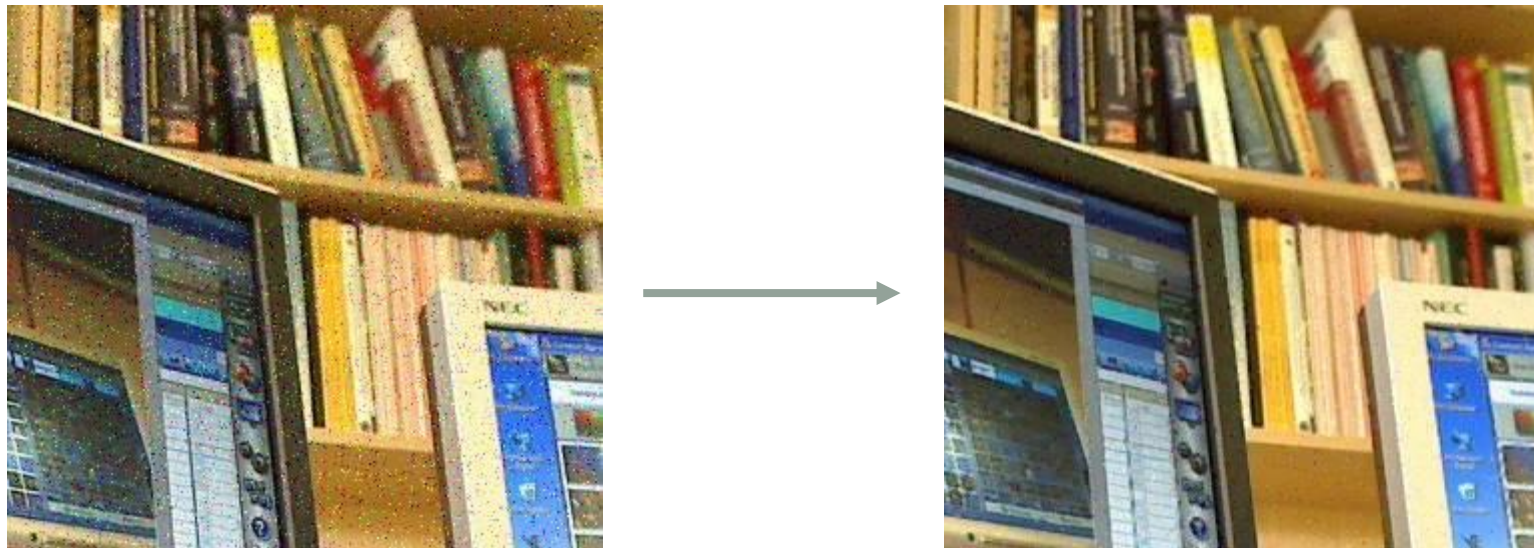
69,632 bytes



5,951 bytes

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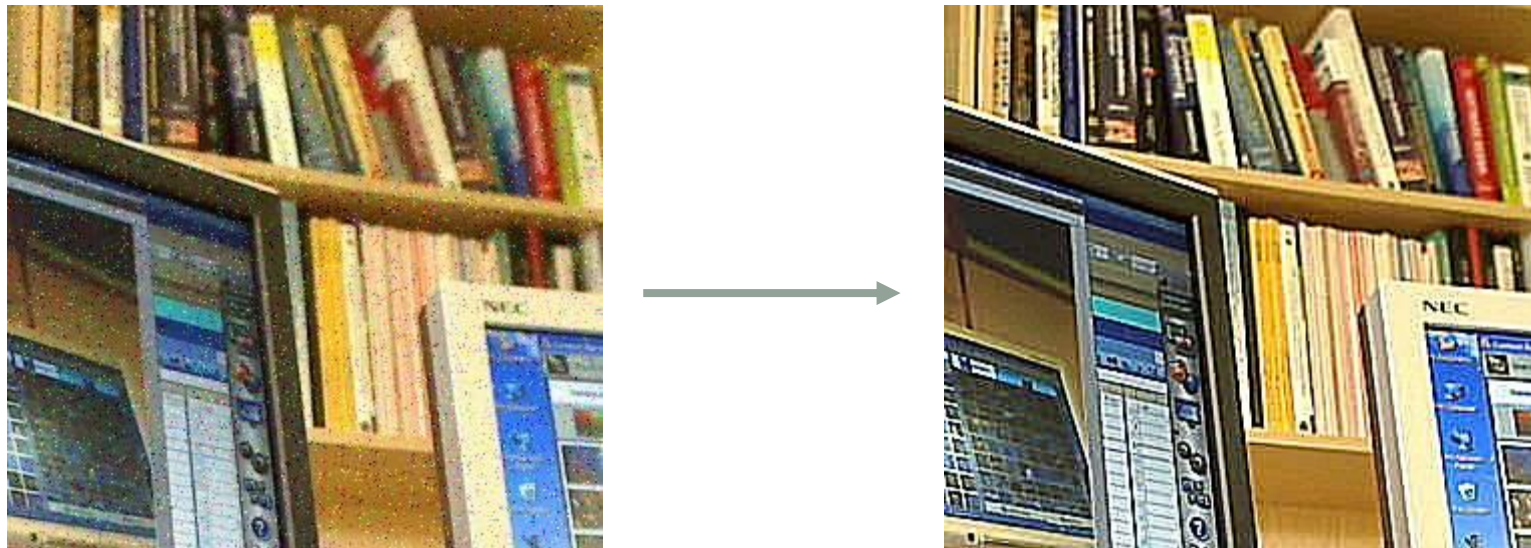
- **Image manipulation** – noise reduction, smoothing, contrast enhancement, etc.



Noise reduction

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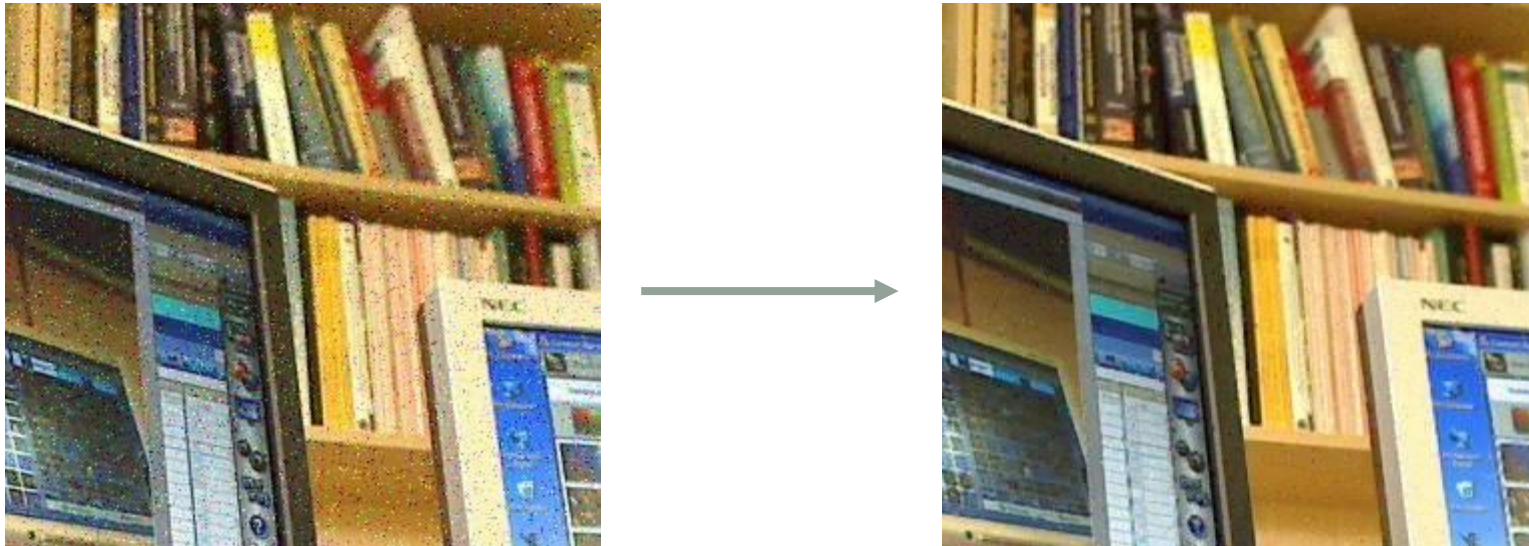
- Image manipulation – noise removal, smoothing, contrast enhancement, etc.



Sharpening

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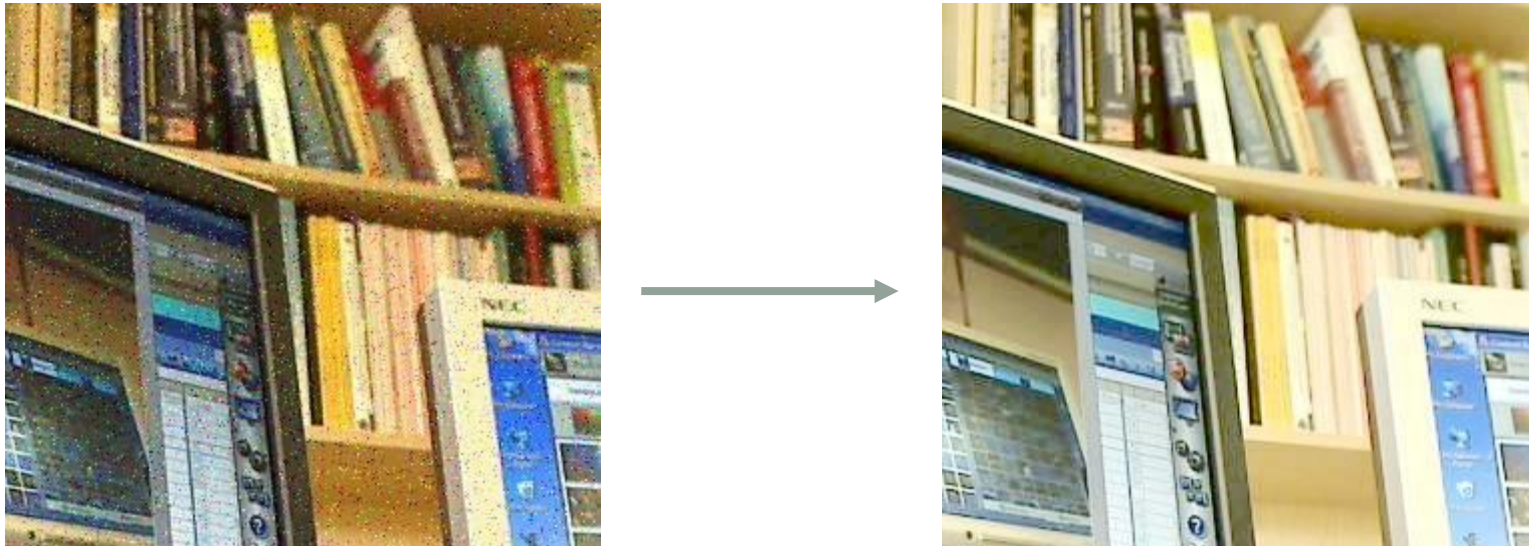
- Image manipulation – noise removal, smoothing, contrast enhancement, etc.



Smoothing

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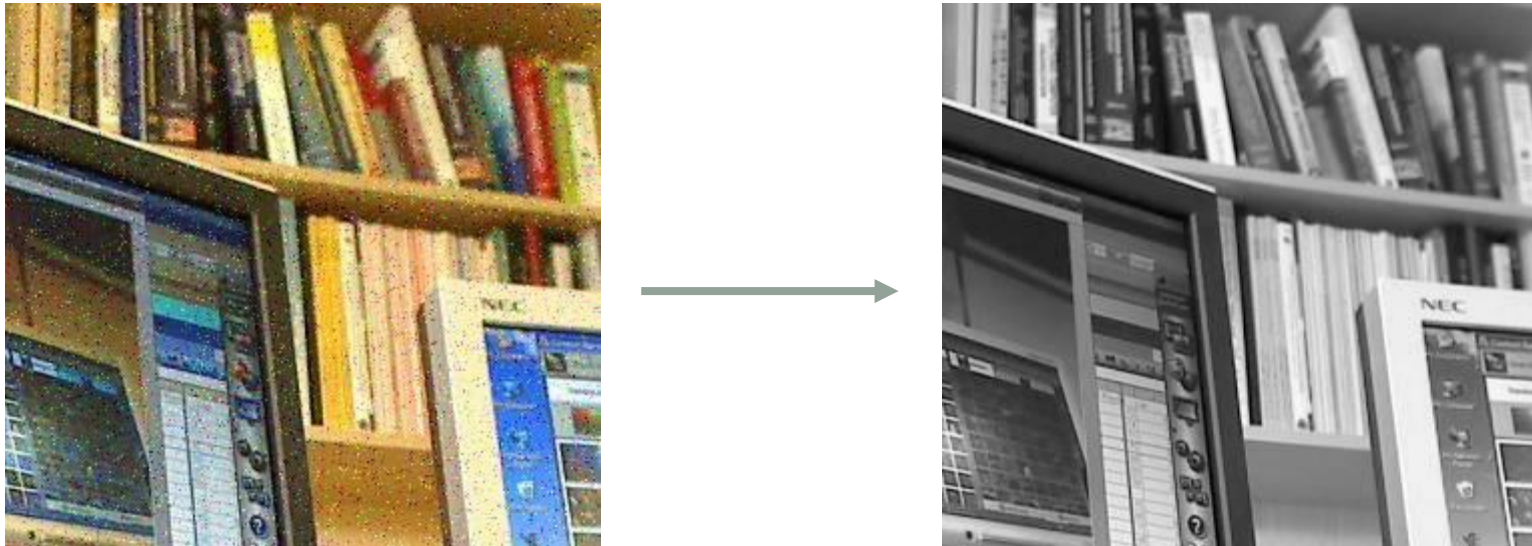
- Image manipulation – noise removal, smoothing, contrast enhancement, etc.



Contrast Enhancement

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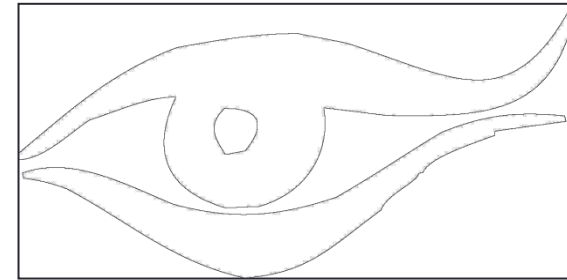
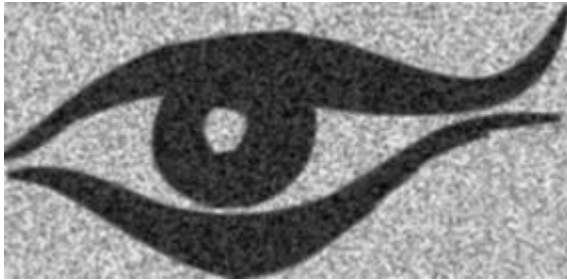
- Image manipulation – noise removal, smoothing, contrast enhancement, etc.



Changing image appearance

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- Edge detection and image segmentation



- Underlying theory
- Some useful algorithms

A step towards image analysis &
computer vision

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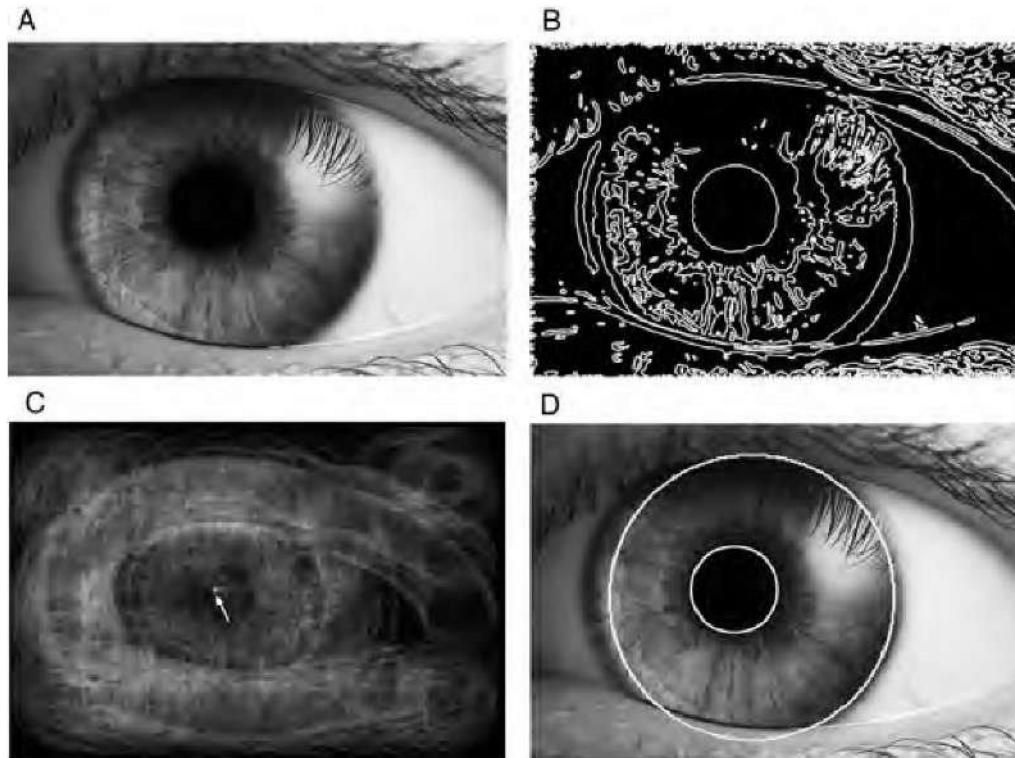
- Superpixels



- an increasingly popular intermediate representation
- fewer data points = less work

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- Finding geometric objects



The Hough Transform

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- Geometric operations – manipulate the array structure underlying the image, not just the pixel values



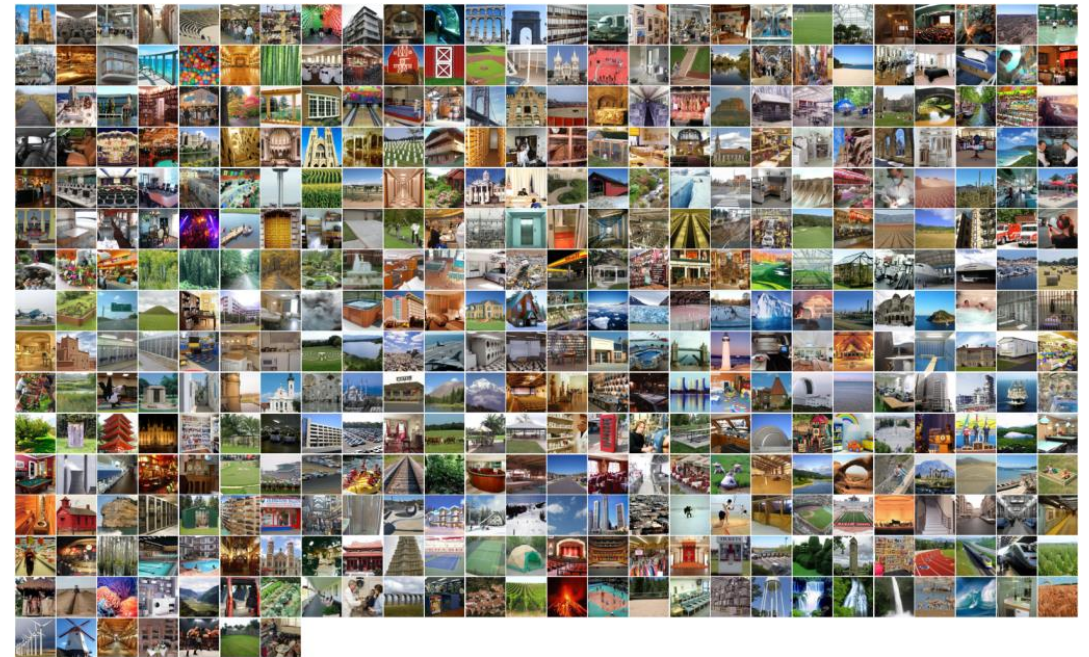
- Rotation, translation, scaling and affine transformations

Some Applications

- Content-based Image Retrieval
- Show me all the images like this...



in here



Some Applications

- Interactive Segmentation



Next Week

Introduction to Point Processing